Service Quality And Performance



Costs of Quality

- A failure to satisfy a customer is considered a defect
- Prevention costs
- Appraisal costs
- Internal failure costs
- External failure costs
- Ethics and quality

Total Quality Management

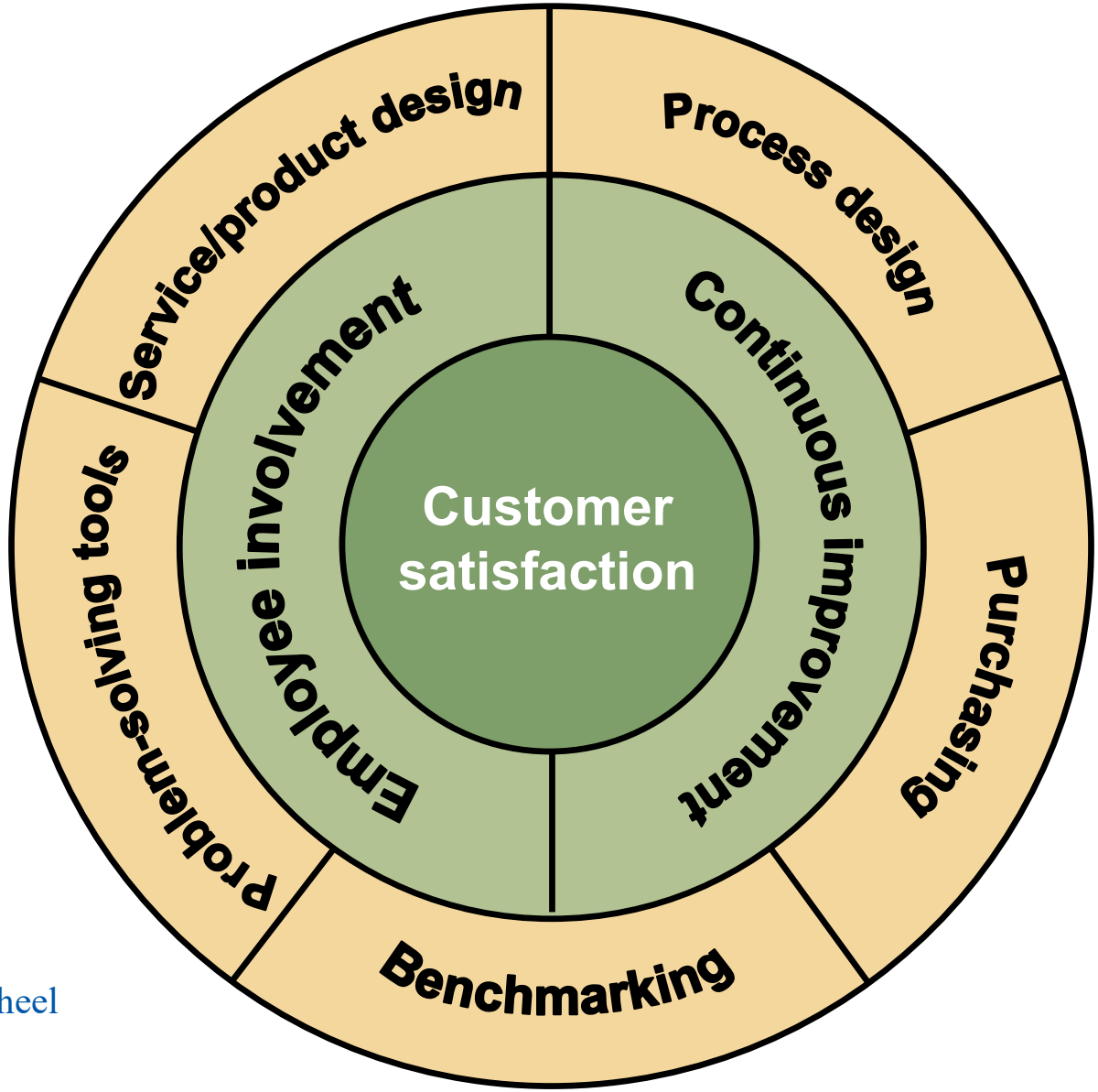


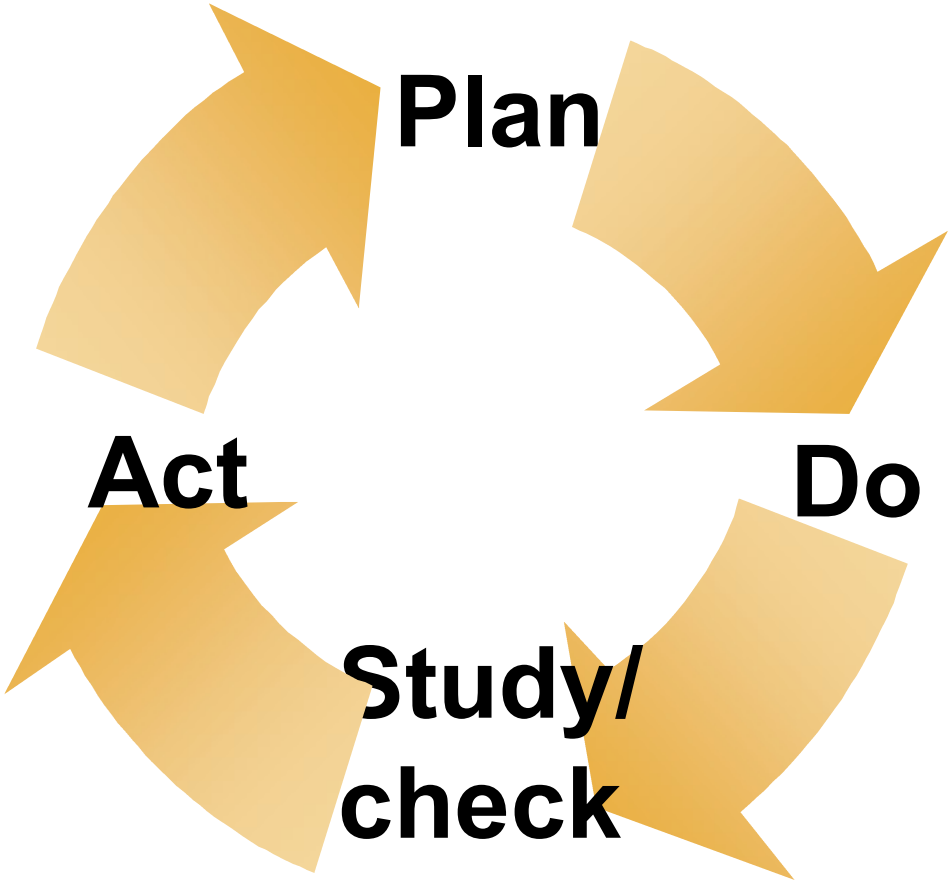
Figure 5.1 – TQM Wheel

Total Quality Management

- Customer satisfaction
  - Conformance to specifications
  - Value
  - Fitness for use
  - Support
  - Psychological impressions
- Employee involvement
  - Cultural change
  - Teams
  - Employee empowerment

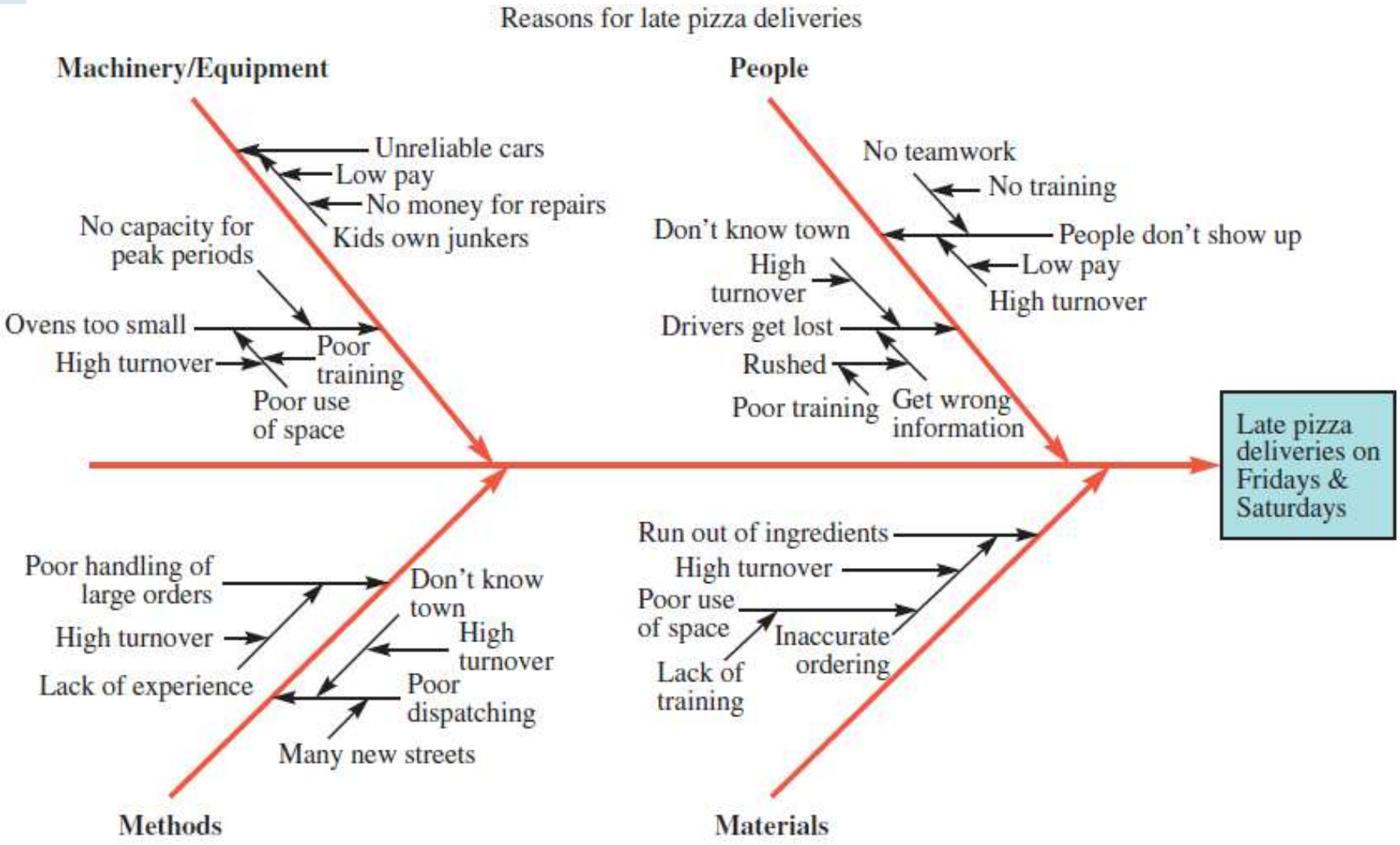
Total Quality Management

- Continuous improvement
  - Kaizen
  - A philosophy
  - Not unique to quality
  - Problem solving process



The Deming Wheel

Cause-and-Effect Diagram (Fishbone Diagram)



Source: From *The Memory Jogger*™II, 2001. Used with permission of GOAL/QPC.

Six Sigma

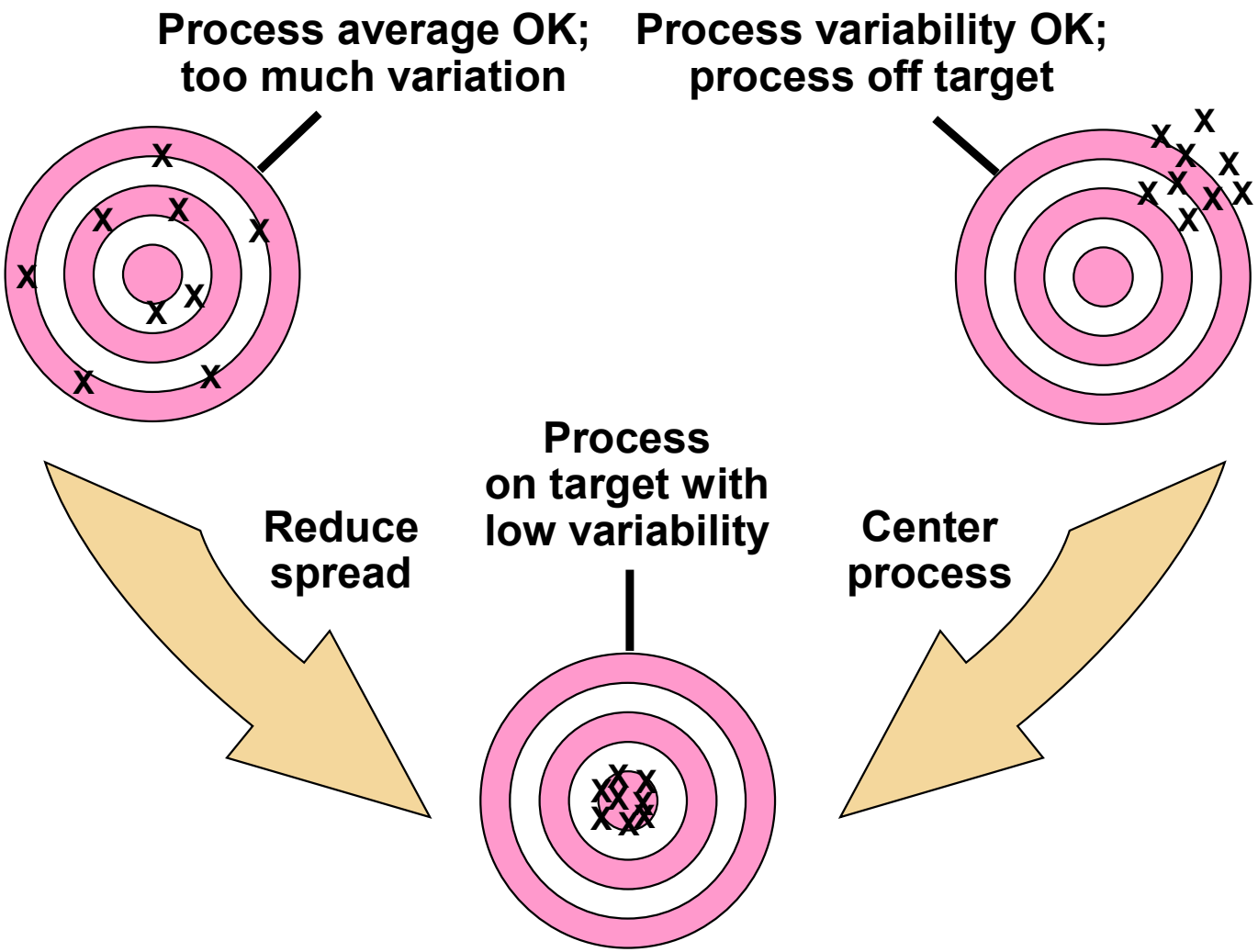


Figure 5.3 – Six-Sigma Approach Focuses on Reducing Spread and Centering the Process

Sampling Distributions

2. The *range* is the difference between the largest observation in a sample and the smallest. The *standard deviation* is the square root of the variance of a distribution. An estimate of the process standard deviation based on a sample is given by

$$s = \sqrt{\frac{\sum (x_i - \bar{x})^2}{n-1}} \quad \text{or} \quad s = \sqrt{\frac{\sum x_i^2 - \frac{\left(\sum x_i\right)^2}{n}}{n-1}}$$

$$E(s^2) = \sigma^2$$

where

s **= standard deviation of a sample**

Motorola design's philosophy

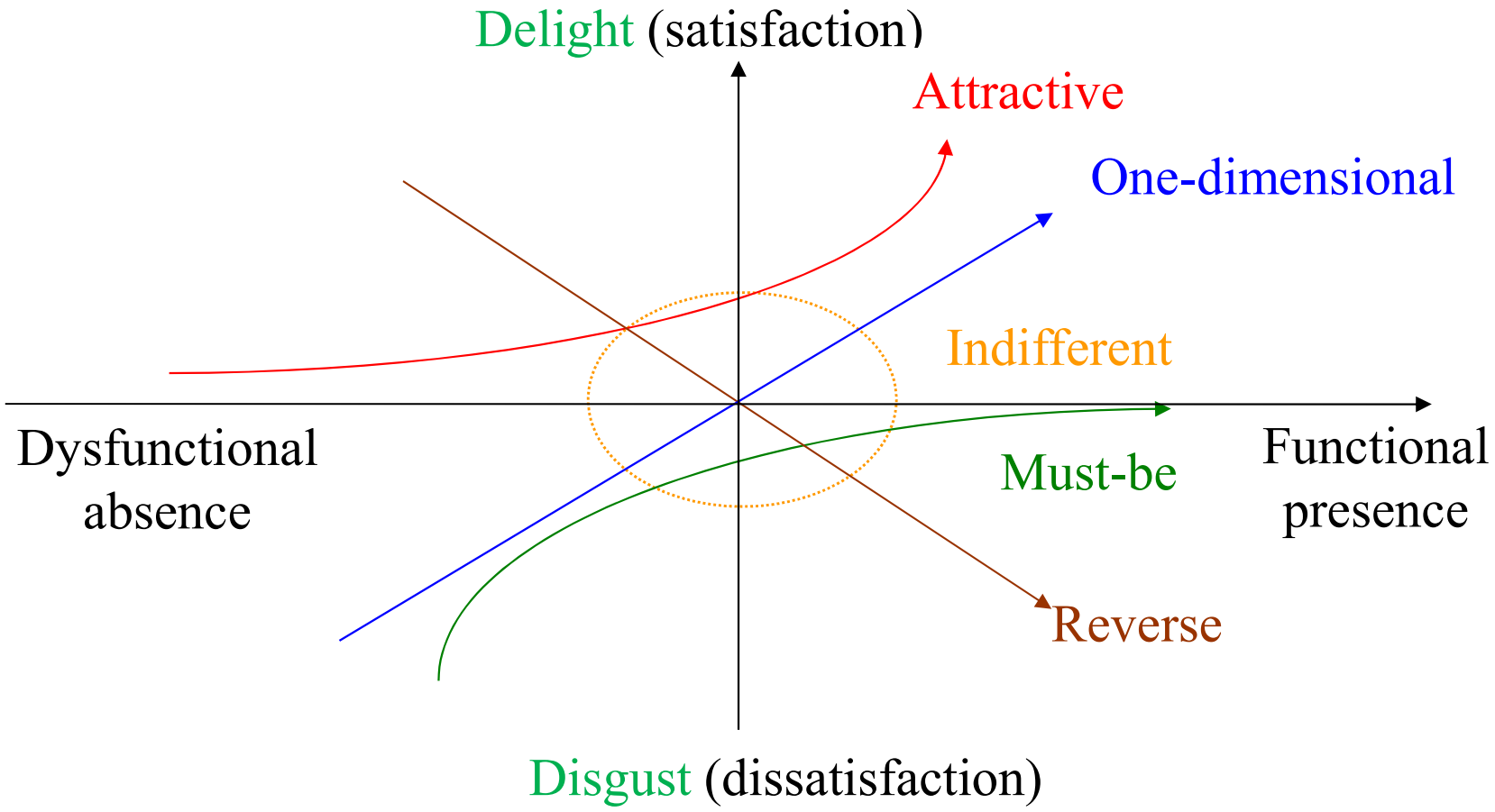
Design limits	C_{pk}	Defective parts	Fraction defective
$\pm 1\sigma$	0.333	317 per thousand	0.3173
$\pm 2\sigma$	0.667	45 per thousand	0.0455
$\pm 3\sigma$	1.0	2.7 per thousand	0.0027
$\pm 4\sigma$	1.333	63 per million	0.000063
$\pm 5\sigma$	1.667	574 per billion	0.000000574
$\pm 6\sigma$	2.0	2 per billion	0.000000002

- Motorola's design specification limit of six sigma with a mean shift by 1.5σ Gives 3.4 defects per million (C_{pk}=1.5). *Why??*

$$C_{pk} = \min\left[\frac{7.5\sigma}{3\sigma}, \frac{4.5\sigma}{3\sigma}\right]$$

$$C_{pk} = \min\left[\frac{\bar{\bar{X}} - LSL}{3\sigma}, \frac{USL - \bar{\bar{X}}}{3\sigma}\right]$$

Concept of Kano Model in service design



Analyze & Brainstorm

- Brainstorming list of features and functionality
- Determining type of product/ service requirements:
- 問卷想了解，屬性存在或不存在時的顧客感受?
- Kano Model Requirements Survey
  - User Survey
    - "Functional form" vs. "Dysfunctional Form"
      - "*How would you feel if the product **had** feature X*?"
      - "*How would you feel if the product **didn't have** feature X*?"
  - Kano Questionnaire Answers:
    - I like it (喜愛).
    - I expect it (期待).
    - I am neutral (沒感覺).
    - I can tolerate it (勉強).
    - I dislike it (厭惡).

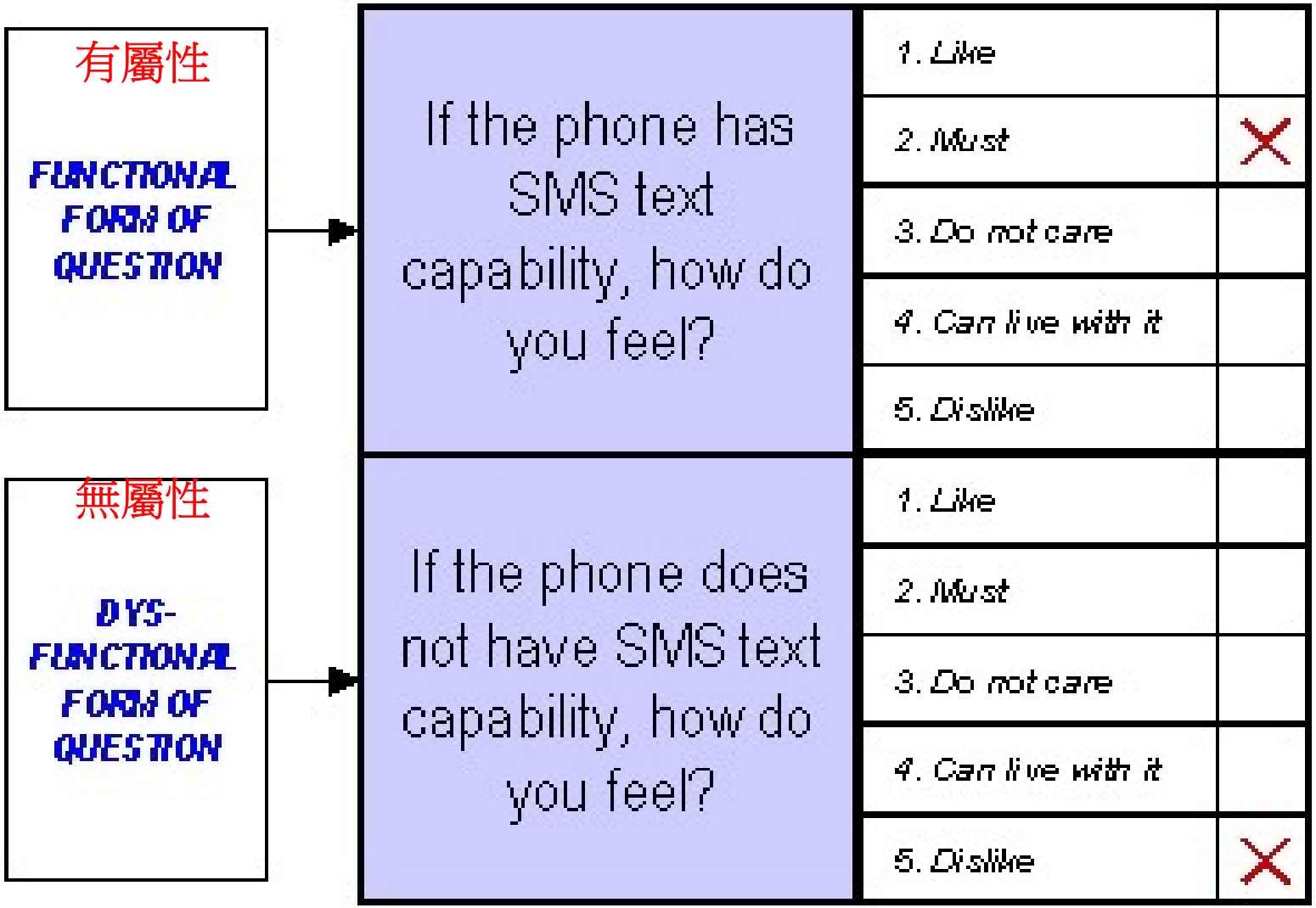
Kano問卷設計

當智慧型手機具備...		滿意	理所當然	沒感覺	能忍受	不滿意
屬性1	正向	*				
	反向				*	
.....	正向					
	反向					
屬性n	正向		*			
	反向					*

問卷問項		負向題(品質要素不充足時)				
		滿意	理所當然	沒感覺	能忍受	不滿意
正向題(品質要素充足時)	滿意	Q	A	A	A	O
	理所當然	R	Q	I	I	M
	沒感覺	R	I	I	I	M
	能忍受	R	I	I	Q	M
	不滿意	R	R	R	R	Q

A代表魅力品質；M代表必需品質；R代表反轉品質；O代表一元品質；Q代表問題品質；I代表無差異品質。

An illustrated Survey



Functional vs. Dysfunctional Comparison

Customer requirements		無屬性DYS-FUNCTIONAL				
		1. Like	2. Must	3. Do not care	4. Can live with it	5. Dislike
有屬性 FUNCTIONAL	1. Like	Q	A	A	A	O
	2. Must	R	I	I	I	M
	3. Do not care	R	I	I	I	M
	4. Can live with it	R	I	I	I	M
	5. Dislike	R	R	R	I	Q

O = One dimensional or linear features
I = Indifferent response
A = Attractive features
E = Expected features
R = reversal (inconsistent response)
Q = Questionable response

M: must feature

Typical airline service attributes

SAs	Definitions of service attributes
A1	Appearance of crew
A2	Courtesy of crew
A3	On-board facilities (seat, space, entertainment, etc.)
A4	On-board beverage and meal
A5	Weight limit of free check-in baggage
A6	Diversity of airline shopping
A7	Weight limit of free carry-on baggage
A8	Fast check-in and boarding process
A9	Speed of handling customer requirements
A10	Skills of handling flight delays or baggage claims
A11	Extended/transferred transportation services
A12	Free services for using the VIP room at airports
A13	Records of low safety accidents
A14	Convenient departure and arrival schedules
A15	Early-bird price discount
A16	Frequent-flier mileage programs

Customer satisfaction based on Kano model

$$d_i^+ = \frac{A_i + O_i - R_i}{A_i + O_i + M_i + R_i + I_i}$$

Positive delight when attribute i is present

$$d_i^- = -\frac{O_i + M_i - R_i}{A_i + O_i + M_i + R_i + I_i}$$

Negative disgust when attribute i is absent

$$S_j = p_j d_j^+ + (1 - p_j) d_j^-$$

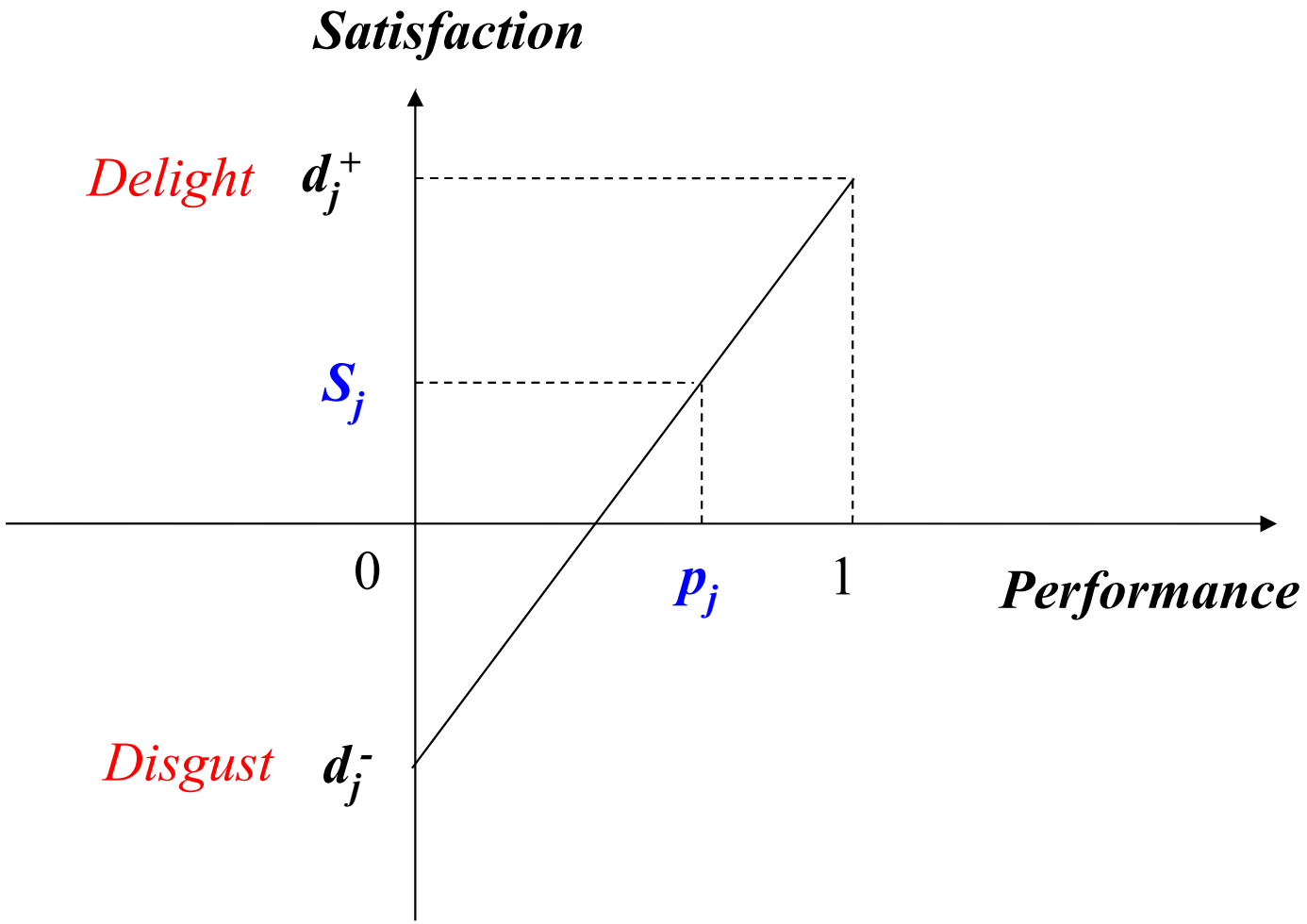
Customer satisfaction with attribute j given performance

績效可由分項分數來計算 (4/5)=0.8

Aggregated Kano categories (in percentages)

SAs	A	M	O	R	I	Q	Delight	Disgust
A1	40%	10%	7%		43%		0.47	-0.17
A2	9%	44%	45%		2%		0.54	-0.89
A3	45%	6%	25%		24%		0.7	-0.31
A4	52%	10%	25%		13%		0.77	-0.35
A5	15%	28%	54%		3%		0.69	-0.82
A6	32%		4%		64%		0.36	-0.04
A7	26%	3%	4%		67%		0.3	-0.07
A8	9%	31%	54%		6%		0.63	-0.85
A9	11%	22%	62%		5%		0.73	-0.84
A10	4%	49%	40%		7%		0.44	-0.89
A11	16%	27%	38%		19%		0.54	-0.65
A12	61%		4%		35%		0.65	-0.04
A13	10%	34%	45%		11%		0.55	-0.79
A14	25%	33%	29%		13%		0.54	-0.62
A15	53%	3%	14%		30%		0.67	-0.17
A16	69%	2%	8%		21%		0.77	-0.1

Converting an attribute's performance into a quantitative degree of satisfaction



Importance-Performance Analysis

